

Lean AI Learning Tools for Philippine Public Schools: Bridging the Gap Between Premium EdTech and Resource Constraints

1 Executive Summary

Current learning tools fail Philippine public schools because they depend on infrastructure that does not exist for most learners: reliable internet (available to only 48.8% of households [1]), personal devices (2.1 million students have none [1]), dedicated IT staff (schools have zero [2]), and teacher bandwidth (55% of teacher time goes to non-teaching tasks [3]). Government platforms like DepEd Commons require connectivity, premium tools like IXL and Khanmigo cost \$20–48 per student annually against a total MOOE of roughly \$50 [4, 5, 6], and DepEd’s own ₱11.4 billion computerization program delivered zero ICT packages in FY 2023 due to procurement failures [1]. The result: 91% learning poverty among late-primary Filipino children [3].

The premium features that actually drive learning gains —adaptive pacing and instant feedback —produce effect sizes of 0.34–0.70 SD across multiple meta-analyses [7, 8, 9], yet they do not require cloud infrastructure or generative AI. Item response theory and spaced repetition algorithms run entirely on-device, and a World Bank meta-analysis found no evidence that generative AI outperforms these older adaptive technologies [10]. Meanwhile, on-device small language models remain premature for this context: Filipino-language performance is poor even for frontier models (GPT-4o scores only 72.23% on Filipino tasks [11]), and most students’ phones lack sufficient RAM.

The highest-impact, most feasible solution is an offline adaptive quiz engine —a self-contained Android APK using IRT-based adaptive difficulty and spaced repetition, aligned to the MATATAG curriculum, requiring no internet, no server, no teacher setup, and no maintenance. At under \$1 per student per year, distributable via Bluetooth or SD card to the low-end Android phones that 98.8% of Filipino internet users already carry [1], it replicates the core mechanism behind the Mindspark RCT’s 0.36 SD math gains [9] and can be prototyped in 12 hours and scaled across 60,000+ public schools. The path to closing the Philippines’ learning gap is not heavier infrastructure but radically simpler software that puts proven adaptive learning directly in each student’s hand.

2 Introduction: The Philippine Learning Crisis

The Philippines faces one of the most severe education crises in the Asia-Pacific region. According to the World Bank’s April 2024 Learning Poverty Brief, approximately 91% of Filipino children at late primary age cannot read and understand a simple text —56 percentage points worse than the East Asia and Pacific regional average [3]. On the 2022 PISA assessment, Filipino students scored 355 in math (ranked 75th of 80 countries), 347 in reading, and 356 in science, with no significant improvement from 2018 [12]. The number of functionally illiterate Filipinos has grown from 14.5 million in 1993 to 24.8 million in 2025, with the national functional literacy rate at just 70.8% per the 2024 FLEMMS [3]. Compared to countries at the same economic level, the Philippines is behind by five to six years’ worth of schooling [3].

This crisis unfolds across more than 60,000 public schools where 2,138,630 learners lack any laptop, desktop, gadget, TV, or radio [1], and only 48.8% of households have internet access [1]. Meanwhile, premium edtech platforms used by private schools demonstrate that adaptive learning, instant feedback, and diagnostic assessment can meaningfully improve outcomes —but at price points and infrastructure requirements that are entirely out of reach for the public system. This report examines why current tools fail, what premium features actually drive learning gains, and how those capabilities can be right-sized into a lean, low-cost, low-maintenance AI learning tool designed for the student as the primary beneficiary —the core challenge of the “Shift to Lean OpenLearning” hackathon track.

3 Current Learning Tools and Why They Fail

3.1 The Tool Landscape

Philippine schools use a mix of government-provided platforms, free commercial tools, and paid subscriptions. DepEd Commons, developed for distance education continuity during the pandemic, reached over 4.2 million users during the Enhanced Community Quarantine period, but sustained post-pandemic adoption data remain limited [3]. The DepEd LMS, a Moodle-based platform, is primarily used for teacher training via NEAP rather than student-facing instruction [13]. Google Classroom became the dominant free LMS post-pandemic across both public and private schools [14], while Facebook and YouTube emerged as the most widely used informal edtech tools due to Facebook’s zero-rated data plans across socioeconomic backgrounds [15]. On the commercial side, Genyo e-Learning by Diwa Learning Systems has been adopted primarily by private schools such as Malate Catholic School and Divine Light Academy [16], while Canvas and Blackboard serve top-tier private universities like De La Salle University, Ateneo de Manila, and the University of Santo Tomas [15].

The flagship DepEd-ADB EdTech project, funded with ₱112 million (including a \$2M ADB grant), equipped only 21 geographically isolated schools with satellite internet, solar power, tablets, and digitized modules —a fraction of the more than 60,000 public schools nationwide [3].

3.2 Why Tools Fail in Public Schools

The failure of these tools in public school settings is driven by five reinforcing barriers. First, connectivity remains deeply unequal: the PSA’s 2024 National ICT Household Survey found that only 48.8% of Filipino households have internet access, dropping to 27.7% in BARMM and just 21.2% in Zamboanga Peninsula [1]. Second, device scarcity is acute —over 2.1 million learners have no digital device of any kind, and among internet users, 98.8% rely on cellphones while only 14.7% have laptops [1]. Third, cost sensitivity is a binding constraint: Philippine schools strongly prefer free or affordable alternatives, and premium EdTech solutions face a constrained market [15]. Fourth, teachers devote only about 45% of their 52-hour work week to actual instruction, with the rest consumed by administrative tasks, leaving little capacity for technology adoption or management [3]. Fifth, language mismatch reduces achievement by 35–57% of learning potential, as most edtech content is in English or Filipino while many students speak different mother tongues [3]. Lower-tier institutions have largely reverted to face-to-face learning post-pandemic, with limited resources making them resistant to sustained technology integration [15].

Tool	Type	Cost/Student/Year	Infrastructure Needs	Management Effort	Key Gap for Public Schools
DepEd Commons	Gov’t platform	Free	Internet required	Low (gov’t-hosted)	Limited post-pandemic adoption; internet-dependent [3]
Google Classroom	Freemium	Free tier	Internet; smartphone	Low-Medium	Requires connectivity; no adaptive learning [14]
Khan Academy	Free	Free	Internet; mobile	Low	Limited adaptive capabilities; English-only [4]
Facebook	Free	Free	Mobile data	None	Unstructured; no learning analytics [15]
YouTube	Social	Free			
Genyo e-Learning	Commercial	School license (paid)	Internet; browser/app	Medium	Cost barrier; private-school focused [16]
Canvas LMS	Commercial	\$5–15/student	Broadband; laptop preferred	High (IT setup)	Too expensive; requires broadband and devices [15]
DepEd-ADB EdTech	Gov’t pilot	~\$95/student (21 schools)	Satellite, solar, tablets	High	Reached only 21 of 60,000+ schools [3]

Table 1: Current Learning Tools in Philippine Schools —Cost, Infrastructure, and Key Gaps

4 What Premium EdTech Delivers and Its Evidence Base

4.1 Premium Platforms and Their Costs

Private schools in the Philippines and across Southeast Asia invest in platforms that offer adaptive learning, AI-powered tutoring, diagnostic assessments, and analytics dashboards. IXL, one of the most widely adopted adaptive platforms, costs approximately \$20–35 per student annually for its comprehensive tier covering math, ELA, science, and social studies, using IRT-based adaptive difficulty and real-time analytics [4]. Mathspace offers step-by-step adaptive feedback on handwritten math work at \$8 (Classes), \$15 (Schools), or \$23 (Schools Plus) per student per year [17]. Khanmigo, Khan Academy’s AI tutor powered by GPT-4, provides Socratic questioning, personalized explanations, and writing coaching at \$4/month (\$48/year) for learners, with a free tier for teachers [5]. Canvas LMS, adopted by the University of Santo Tomas system and De La Salle Zobel, offers full LTI integrations and analytics at custom institutional pricing [18]. Quipper, a Philippines-focused platform, provides LMS functionality with content delivery, assessment tools, and progress tracking at institutional pricing [19].

4.2 Evidence That Premium Features Improve Learning

The four features most consistently linked to measurable learning gains are personalized adaptive pacing, instant step-level feedback, diagnostic assessments, and teacher analytics dashboards. A 2024 meta-analysis in the *Journal of Educational Computing Research* found AI-enabled adaptive learning systems produce a medium-to-large effect size ($g = 0.70$) on cognitive learning outcomes compared to non-adaptive interventions [7]. A 2026 World Bank meta-analysis drawing on 191 effect sizes from 14 RCTs across 10 economies found that adaptive and AI-enabled interventions raise student learning by 0.125 SD relative to traditional instruction [20]. A 2024 meta-analysis on AI in K-12 mathematics found an overall effect size of 0.343, with AI showing greater impact when serving as an intelligent tutoring system or adaptive learning system [8]. A 2026 *Frontiers in Psychology* meta-analysis found personalized AI systems in blended learning had the highest (large) effect compared to chatbots and intelligent tutoring systems alone [21]. Perhaps most relevant for the Philippine context, the Mindspark RCT in India demonstrated that adaptive computer-based instruction increased math scores by 0.36 σ and Hindi scores by 0.22 σ after just 4.5 months of access [9].

Platform	Cost/Student/Year	Adaptive Learning	AI Tutoring	Analytics	Evidence of Impact
IXL	\$20–35 [4]	Yes (IRT-based)	No	Advanced diagnostics	Meta-analyses: 0.12–0.70 SD [7, 20]
Comprehensive Mathspace	\$15–23 [17]	Yes (step-level)	No	Progress tracking	Adaptive feedback linked to gains [8]
Schools Khanmigo	\$48 [5]	Yes	Yes (Socratic)	Moderate	Personalized AI: large effect [21]
AI Khan Academy	\$0 [4]	Limited	No	Basic	Less structured than premium [4]
(Free) Kolibri	\$0 (self-hosted)	Limited	No	Coach dashboard	Deployed in 195+ countries [22]
(Open-source)					

Table 2: Premium vs. Free EdTech Platforms —Features, Costs, and Evidence

The cost gap is stark: the features that drive the strongest learning outcomes —adaptive pacing and instant feedback —are locked behind \$8–48/student/year subscriptions, while free alternatives like Khan Academy offer extensive content but limited adaptive learning capabilities and less structured progression [4].

5 Public School Constraints: The Reality Any Solution Must Fit

5.1 Classroom and Staffing Realities

Philippine public school classrooms are severely overcrowded. National averages are 32 students per classroom at the elementary level and 44 at junior high, but congested regions such as NCR, CALABARZON, and BARMM reach 60 students per classroom [3]. Teachers spend only 45% of their 52-hour work week on actual teaching, with 55% consumed by non-teaching administrative tasks; DepEd orders to reduce this workload have seen uneven implementation [3]. In SY 2022–2023, 5,870 public schools (about 12% of all public schools) offered multigrade classes, serving 277,908 learners [1]. Schools lack dedicated IT staff entirely —DepEd policy directs schools to designate existing teachers as ICT Coordinators, with two for large/medium schools and one for small schools [2], meaning technology maintenance falls on already overburdened educators.

5.2 Connectivity, Electricity, and Devices

The digital infrastructure gap is severe. Among multigrade schools, 51% have no internet access and 15% lack electricity entirely [1]. Approximately 1,500 public schools still lack access to electricity as of October 2024 [1]. A DepEd survey found that 2,138,630 learners have no laptop, desktop, gadget, TV, or radio; among individual internet users, cellphones remain the dominant device at 98.8%, while only 14.7% use laptops and 10.0% use desktops [1]. Any viable solution must assume the student’s device is a low-end Android phone —or no device at all.

5.3 Budget and Procurement Failures

The MOOE (Maintenance and Other Operating Expenses) per learner in elementary public schools has risen from ₱1,389 in 2024 to ₱2,140 in 2025 to ₱2,792 in 2026 [6] —roughly \$50 per student per year at current exchange rates, covering all school operating costs, not just technology. The DepEd Computerization Program had a FY 2023 budget of approximately ₱11.4 billion but delivered zero ICT packages from current funds, achieving a 0% target completion rate due to severe procurement delays [1]. A separate laptop procurement scandal saw ₱979 million in overpricing and ₱667 million worth of laptops never delivered to teachers [16]. These failures demonstrate that any solution dependent on centralized government procurement is high-risk.

5.4 Language Diversity

The Philippines has over 180 languages, but only 19 have formal MTB-MLE support including official orthography, learning resources, and teacher training materials. A survey found that only 9% of 16,287 schools completed all four required MTB-MLE activities [3]. Republic Act 12027 (2024) shifted from mandatory to conditional MTB-MLE, defaulting to Filipino and English instruction [3], but the language barrier remains a fundamental obstacle to learning —lessons delivered in a mismatched language reduce achievement by 35–57% of learning potential [3].

6 Lean Deployment Approaches and Right-Sizing Opportunities

6.1 Which Premium Features to Keep and Which to Cut

The evidence is clear that adaptive pacing and instant feedback are the highest-value features, consistently producing the largest effect sizes across meta-analyses [7, 8, 21]. These can be delivered through lightweight algorithms (item response theory, spaced repetition) that run entirely on-device without cloud infrastructure. Diagnostic assessments can be approximated through pre-loaded question banks with Bayesian scoring. The features that should be cut for public school deployment are real-time teacher analytics dashboards (teachers lack time to use them), cloud-dependent AI tutoring (requires connectivity), and rich multimedia content delivery (requires bandwidth and storage).

Critically, a World Bank meta-analysis found no evidence that generative AI tools outperform the older adaptive technologies that preceded them [10], and a systematic review in *npj Science of Learning* found that AI-driven intelligent tutoring system effects are “generally positive but mitigated when compared to non-intelligent tutoring systems”[23]. This means well-designed non-AI adaptive tools —spaced repetition, IRT-based quizzing, pre-loaded content —can match AI tutors for drill, practice, and formative assessment at a fraction of the cost and complexity.

6.2 Lean Deployment Architectures

Chart Explanation: This chart compares the annual per-student cost of premium platforms (red), free cloud-based tools (blue), and lean offline solutions (green) against the Philippine public school MOOE per learner of approximately \$50/year [6]. Lean solutions cluster below \$5/student, while premium platforms consume a significant share of the total operating budget.

Approach	Cost/Student/Year	Maintenance Burden	Main Risk	Proven Deployment
Offline-first apps (Kolibri)	~\$2–5 (software free; shared hardware)	Low–Medium: initial setup, periodic content updates	Content staleness; device breakage [24]	195+ countries; 8 schools in rural Zambia (2024) [24, 22]
SMS/USSD delivery (Eneza model)	~\$2–5/year (~\$0.04/day)	Very low: centrally managed	160-char limit; carrier dependency [25]	Kenya: 22.7% higher scores [26]; EdTech Hub confirmed positive impact [27]
Raspberry Pi edge servers (RACHEL/ LEAP)	~\$3–8 (hardware shared across 40 students)	Medium: periodic content refresh via USB	Hardware theft; single point of failure; power supply [28]	Kiribati (110 schools) [29]; India (LEAP with AID India) [28]

Approach	Cost/Student/Year	Maintenance Burden	Main Risk	Proven Deployment
On-device SLMs (0.5B–1.5B params)	\$2–4 (requires mid-range phone with 3GB+ RAM)	High: model updates, hallucination monitoring	Needs phones most students lack; weak Filipino performance [30, 11]	No proven education deployments at scale
Hybrid sync-when-online	~\$5–10 (RPI + periodic connectivity)	Medium: manifest management, USB sneakernet	Sync failures; version conflicts [28]	LEAP platform in India [28]

Table 3: Lean Deployment Approaches —Cost, Maintenance, and Risk

Kolibri, developed by Learning Equality, is a free, open-source offline-first platform deployed across 195+ countries, designed specifically for low-resource environments with limited teacher capacity and large class sizes [22]. In January 2024, Impact Network launched a pilot in eight community schools in rural Zambia [24]. **Eneza Education’s Shupavu291** in Kenya delivers SMS-based learning aligned to the national curriculum at approximately \$0.04/day per student, and UNESCO reports that students using Eneza score 22.7% higher than their peers [25, 26]. **LEAP** (Low-bandwidth Educational Access Platform) uses a Raspberry Pi 4 with USB storage and a router to serve approximately 40 students per node, with content synced from AWS S3 or via USB sneakernet for zero-bandwidth sites [28]. **RACHEL** servers by World Possible run on Raspberry Pi with pre-loaded Khan Academy and Wikipedia content, deployed across 110 schools in Kiribati [29].

6.3 On-Device Small Language Models: Promising but Premature

Running quantized small language models on phones is technically feasible but faces significant constraints. Tiny-MobileLLM benchmarks show a Qwen 0.5B model achieves approximately 16 tokens/sec using 852 MB RAM on a Snapdragon 855 device, while a 1.5B model requires about 1,290 MB [30]. However, most Philippine public school students have phones with 2–3GB RAM, making only the smallest models viable. More critically, Filipino language quality is a major gap: the FilBench evaluation of 27 LLMs found that even the best model (GPT-4o) scored only 72.23% on Filipino tasks [11], and small on-device models will perform substantially worse. A scoping review confirms that low-resource settings should prioritize basic access tools over personalized AI systems [31]. For a hackathon prototype, a 0.5B model could handle structured educational Q&A (multiple-choice generation, hint provision) but should not be relied upon for open-ended Filipino generation.

7 Candidate Solution Directions

Based on the constraints and evidence above, five student-facing solution directions emerge, ranked by combined learning impact, 12-hour hackathon prototype feasibility, and scalability across 60,000+ public schools.

Chart Explanation: This chart scores each candidate solution across three dimensions on a 1–5 scale. The Offline Adaptive Quiz Engine achieves the highest combined score (14/15) due to its strong scalability via simple APK distribution, high hackathon feasibility using open-source IRT libraries, and proven learning impact from adaptive pacing evidence [9, 7].

7.1 Solution 1: Offline Adaptive Quiz Engine (Score: 14/15)

This solution addresses the core problem that 91% of Filipino children at late primary age cannot read and understand a simple text [3], largely because classroom instruction cannot personalize pace for 44–60 students [3]. The engine uses pre-loaded question banks aligned to the MATATAG curriculum with Bayesian item-response theory scoring running entirely on-device. Spaced repetition ensures students revisit weak areas automatically. It runs as a self-contained Android APK requiring no internet, no server, and no teacher setup. Estimated cost is under \$1 per student per year (open-source, content-only cost). Teachers and schools need to do nothing to maintain it—the APK is distributed via Bluetooth, SD card, or a single download. The Mindspark RCT demonstrated that this type of adaptive computer-based instruction can increase math scores by 0.36σ after just 4.5 months [9]. This is the top recommendation for the hackathon: it can be built in 12 hours using open-source IRT libraries and scales to 60,000+ schools via APK distribution.

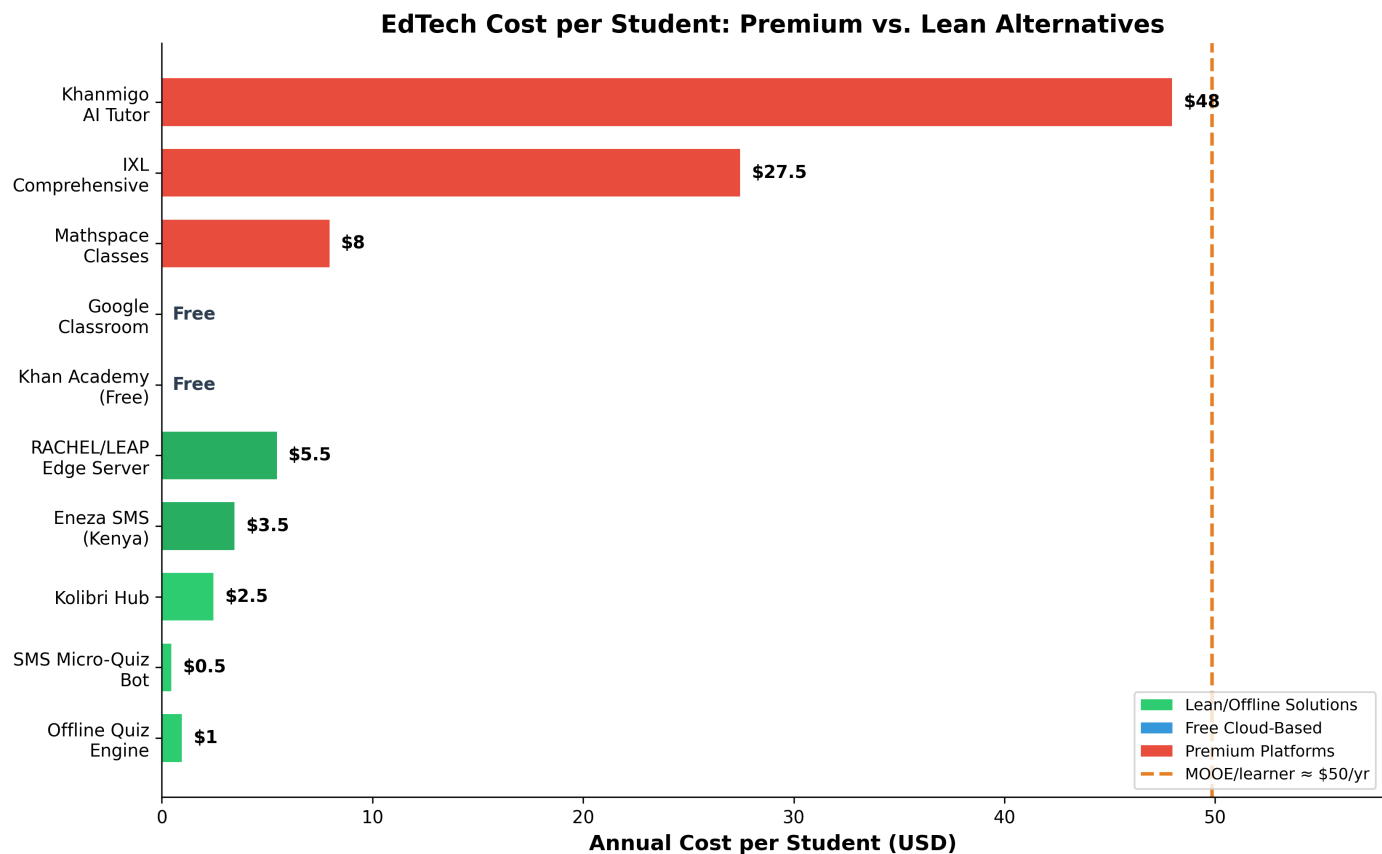


Figure 1: EdTech Cost per Student: Premium vs. Lean Alternatives

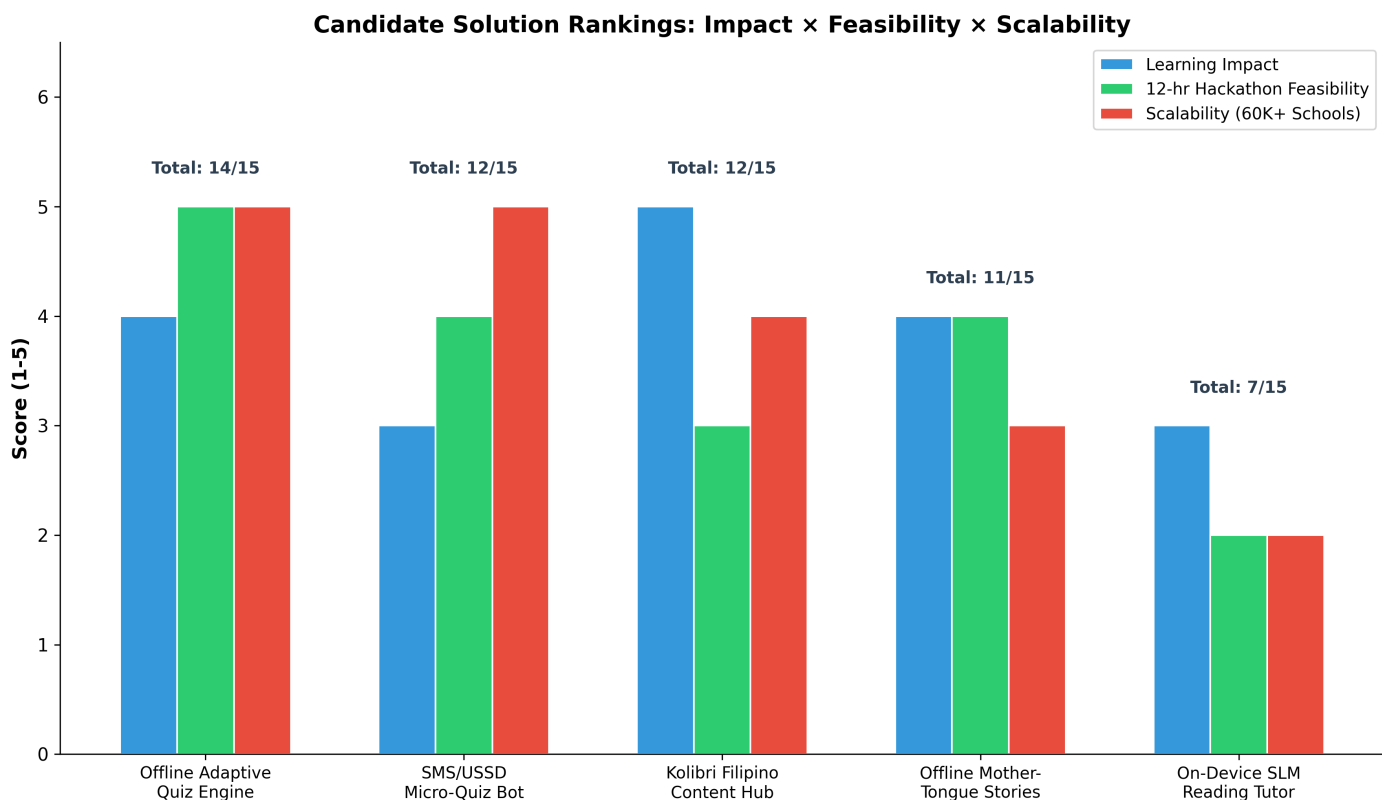


Figure 2: Candidate Solution Rankings

7.2 Solution 2: SMS/USSD Micro-Quiz Bot (Score: 12/15)

This solution targets the 2.1 million learners who lack any smart device [1] but may have access to a basic feature phone. It delivers daily 3-question math and literacy quizzes via SMS or USSD, with adaptive difficulty managed server-side. Modeled on Kenya's Eneza Education, which costs approximately \$0.04/day and produces 22.7% higher scores [25, 26], this approach requires a telco partnership (e.g., Globe or Smart in the Philippines) for zero-rated SMS. Cost is under \$0.50 per student per year with a telco subsidy. Teachers do nothing—the system is centrally managed. The main limitation is the 160-character constraint, which restricts content depth and eliminates rich media.

7.3 Solution 3: Kolibri Filipino Content Hub (Score: 12/15)

This solution leverages the proven Kolibri offline-first platform [22] with curated Filipino and mother-tongue K–3 reading and math content, addressing the language barrier that reduces achievement by 35–57% [3]. It runs on a shared school tablet or phone with content pre-loaded, requiring no internet for daily use. Cost is approximately \$2–3 per student per year (amortized device and content costs). A teacher uses the built-in coach dashboard for basic monitoring, but the platform is designed for minimal teacher capacity [22]. The main challenge is initial content curation in Filipino and regional languages, and the need for at least one shared device per classroom.

7.4 Solution 4: Offline Mother-Tongue Story Generator (Score: 11/15)

This solution directly addresses the fact that only 19 of 180+ Philippine languages have formal MTB-MLE support [3]. It provides a pre-generated bilingual story library (Filipino plus a regional language) with comprehension exercises and rule-based adaptive difficulty. Content is static and packaged as an APK, requiring no internet or AI inference. Cost is under \$1 per student per year. The limitation is that content must be pre-generated for each target language, reducing scalability to new language communities without additional content development effort.

7.5 Solution 5: On-Device SLM Reading Tutor (Score: 7/15)

This solution uses a quantized 0.5B parameter model to provide Filipino reading comprehension Q&A and pronunciation feedback via text-to-speech. While technically feasible on phones with 3GB+ RAM at approximately 16 tokens/sec [30], it ranks last due to three critical constraints: most public school students lack phones with sufficient RAM [1], Filipino language performance is poor even for the best LLMs (GPT-4o scores only 72.23% on Filipino tasks) [11], and sustained inference causes thermal and battery issues. Cost is \$2–4 per student per year, but the device requirement effectively excludes the most disadvantaged learners. This approach is better suited as a future enhancement once device penetration improves and Filipino-optimized small models mature.

8 Conclusion: The Single Biggest Opportunity

The single biggest opportunity for the “Shift to Lean OpenLearning” hackathon is an **offline adaptive quiz engine**—a self-contained Android APK that uses item response theory and spaced repetition to deliver personalized math and reading practice aligned to the MATATAG curriculum, requiring no internet connection, no cloud infrastructure, no teacher setup, and no IT maintenance. This approach directly targets the 91% learning poverty rate [3] by meeting each student at their actual skill level rather than the class average, replicating the adaptive pacing feature that drives the largest effect sizes in premium platforms (0.36 σ in the Mindspark RCT [9], up to 0.70 in meta-analyses [7]) at under \$1 per student per year. It can be prototyped in 12 hours using open-source IRT libraries, distributed to 60,000+ schools via Bluetooth or SD card sharing, and used by any student with access to even the most basic Android phone—the device that 98.8% of Filipino internet users already rely on [1]. In a system where ₱11.4 billion in ICT budget delivered zero packages [1] and premium platforms cost \$20–48 per student [4, 5], the path forward is not more infrastructure or more spending—it is radically simpler software that puts adaptive learning directly in the student's hand.

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